4 COMBINATORICS

4.1 Introduction

Let $m \in \mathbb{N}$. For a procedure of m successive distinct and independent steps with n1 outcomes possible for the first step, n_2 outcomes possible for the second step, ... , and nm outcomes possible for the mth step, the total number of possible outcomes is

$$n_1 \cdot n_2 \cdot \cdot \cdot n_m$$

For a collection of m disjoint sets with n_1 elements in the first, n_2 elements in the second, . . ., and nm elements in the mth, the number of ways to choose one element from the collection is

$$n_1 + n_2 + \cdot \cdot \cdot + n_m$$

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{n-k} x^k y^{n-k}.$$

$$\sum_{i=0}^{n} C(n,i)(-1)^i = 0$$

For n even,

$$\sum_{i=0}^{n/2} C(n, 2i) = \sum_{i=0}^{n/2-1} C(n, 2i+1)$$

For n odd,

$$\sum_{i=0}^{\lfloor n/2 \rfloor} C(n, 2i) = \sum_{i=0}^{\lfloor n/2 \rfloor+1} C(n, 2i+1)$$

$$\sum_{i=1}^{n} iC(n,i) = n2^{n-1}.$$

For positive integers n, t, the coefficient of $x_1^{n_1} x_2^{n_2} x_3^{n_3} \ldots x_t^{n_t}$ in the expansion of $(x_1 + x_2 + x_3 + \ldots + x_t)^n$ is

$$\frac{n!}{n_1!n_2!n_3!\ldots n_t!},$$

where each n_i is an integer with $0 \le n_i \le n$, for all $1 \le i \le t$, and $n_1 + n_2 + n_3 + \cdots + n_t = n$.

When we wish to select, with repetition, r of n distinct objects, we are considering all arrangements of r x's and n – 1 |'s and that their number is

$$\frac{(n+r-1)!}{r!(n-1)!} = \binom{n+r-1}{r}.$$

Consequently, the number of combinations of n objects taken r at a time, with repetition, is C(n + r – 1, r).

4.1.1 we recognize the equivalence of the following:

(a) The number of integer solutions of the equation

$x_1 + x_2 + \ldots + x_n = r,\ x_i \geq 0,\ 1 \leq i \leq n.$

(b) The number of selections, with repetition, of size r from a collection of size n.

(c) The number of ways r identical objects can be distributed among n distinct containers.

$$b_n = \binom{2n}{n} - \binom{2n}{n-1} = \frac{1}{n+1}\binom{2n}{n}, \qquad n \geq 1, b_0 = 1.$$

The numbers $b_0, b_1, b_2, \ldots$ are called the Catalan numbers, after the Belgian mathematician

Order Is Relevant	Repetitions Are Allowed	Type of Result	Formula
Yes	No	Permutation	$P(n, r) = n!/(n-r)!,\ 0 \leq r \leq n$
Yes	Yes	Arrangement	$N^r,\ n, r \geq 0$
No	No	Combination	$C(n, r) = n!/[r!(n-r)!] = \binom{n}{r}$, $0 \leq r \leq n$
No	Yes	Combination with repetition	$\binom{n+r-1}{r}$, $n, r \geq 0$

If there are n objects with n_1 indistinguishable objects of a first type, n_2 indistinguishable objects of a second type, . . . and n_r indistinguishable objects of an rth type, where $n_1 + n_2 + \ldots + n_r = n$, then there are $\frac{n!}{n_1!n_2!\ldots n_r!}$ (linear) arrangements of the given n objects.

Let A and B be subsets of a finite universal set U. Then

(a) $|A \cup B| = |A| + |B| - |A \cap B|$

(b) $|A \cap B| \leq \min\{|A|, |B|\}$, the minimum of |A| and |B|

(c) $|A \backslash B| = |A| - |A \cap B| \geq |A| - |B|$

(d) $|A^c| = |U| - |A|$

(e) $|A \oplus B| = |A \cup B| - |A \cap B| = |A| + |B| - 2|A \cap B| = |A / B| + |B / A|$

(f) $|A \times B| = |A| - |B|$